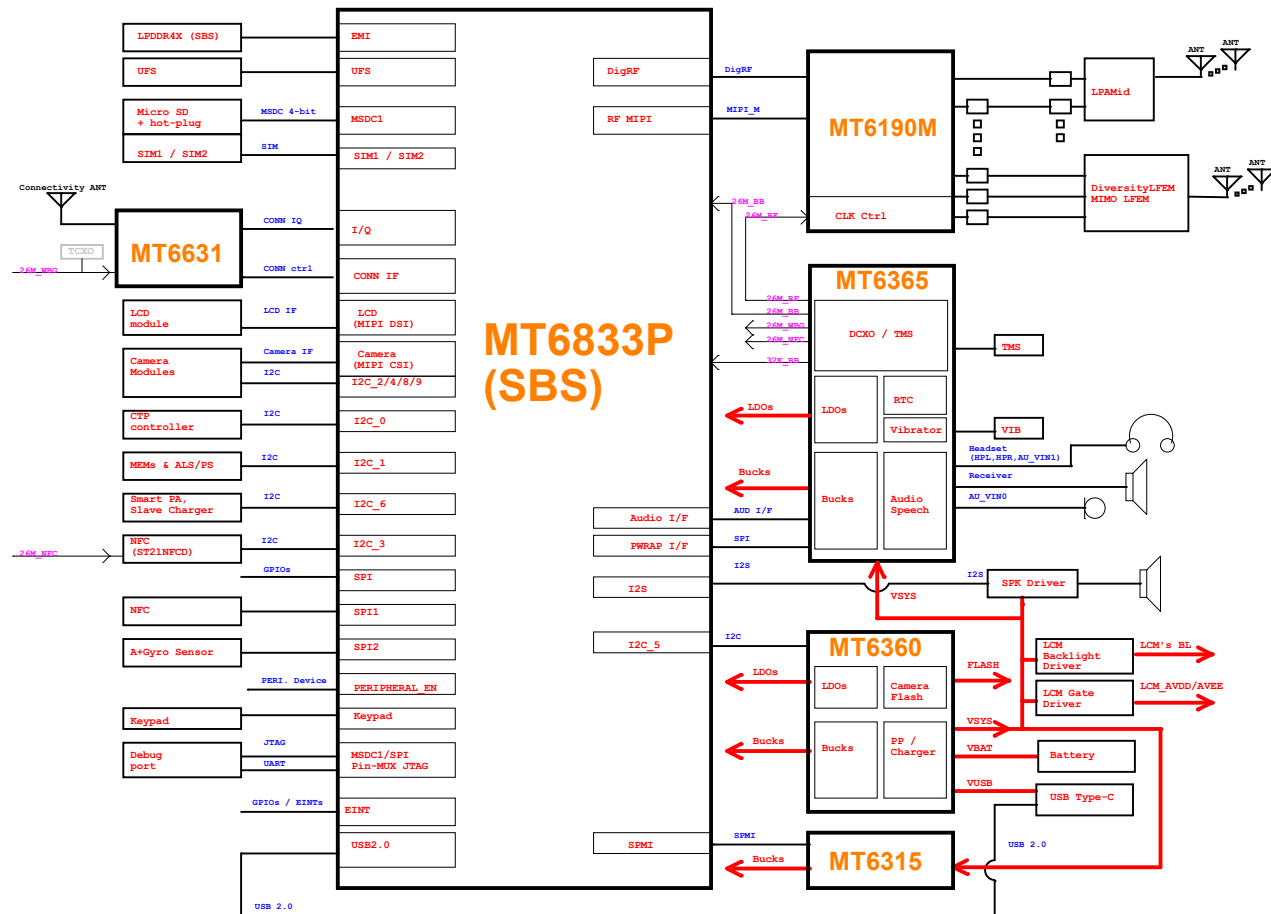




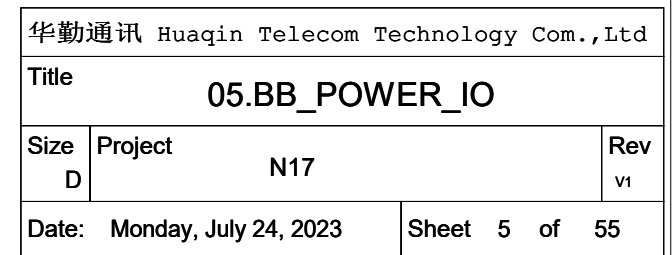
华勤通讯 Huaqin Telecom Technology Com., Ltd		
Title Block Diagram		
Size D	Project N17	Rev V1
Date: Monday, July 24, 2023		Sheet 1 of 55

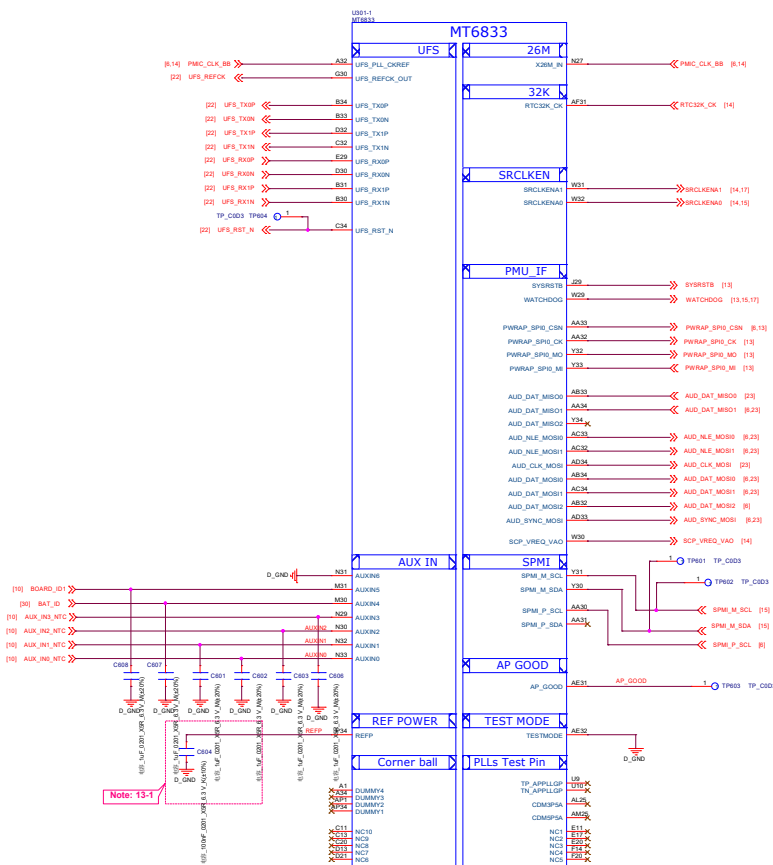
Date	Version	Page	Change notice
2020.06.xx	Pre-Released		
2020.07.xx	V01		
2020.08.14	V02	13_BB_1 29_POWER_EXT 20_POWER_MT6365_Buck 25_POWER_MT6315_VMD	Disconnect Dummy pins A1,A34,AP1,AP34 to GND Change C2923 cap from 6.3V to 10V [PDN] Change Caps of DVDD_GPU - C2039/C2038 to NC [PDN] Change Caps of DVDD_MODEM - C2502/C2534/C2523/C2522 to NC
2020.08.26	V03	File name	Modify the file name

华勤通讯 Huaqin Telecom Technology Com.,Ltd		
Title Change Notice		
Size D	Project N17	Rev V1
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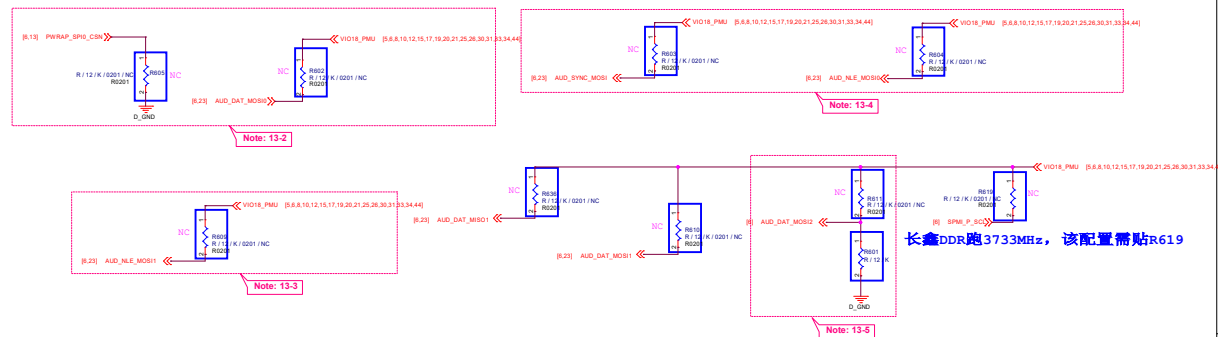
华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title 03.BB_POWER_PDN1			
Size D	Project N17		Rev v1
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Note 13-6: "AUD_DAT_MISO1" is trapping pin to select VEMC Voltage.

AUD_DAT_MISO1	VEMC Voltage
L (Default)	VEMC=3.0V for VCC of UFS2.* device
H (by external PU)	VEMC=2.5V for VCC of UFS3.0 device



Schematic design notice:

Note 13-1: The load cap. have to be placed as close to REFP ball as possible.

Note 13-2: "PWRAP_SPIO_CSN" and "AUD_DAT_MOSI0" pin features in trapping pin to enable JTAG.

PWRAP_SPIO_CSN	AUD_DAT_MOSI0	AP JTAG	IO JTAG
H (Default)	L (Default)	N/A	N/A
L (by external PD)	L	SPI0+EINT8	EINT16/EINT17/EINT18/EINT19/EINT20/EINT21/EINT22/IC7
L (by external PD)	H (by external PU)	MSDC1_CLK, MSDC1_CMD, MSDC1_DAT0, MSDC1_DAT1, MSDC1_DAT2	N/A

Note 13-3: "AUD_NLE_MOSI1" features in trapping pin to enable JTAG over USB 2.0 port.

AUD_NLE_MOSI1	USB JTAG
L (Default)	Disable serial JTAG over USB
H (by external PU)	DP/IDM output JTAG

Note 13-4: "AUD_SYNC_MOSI" and "AUD_NLE_MOSI0" pin features in trapping pin to booting (eMMC/UFS/SPI NOR).

AUD_SYNC_MOSI	AUD_NLE_MOSI0	Storage Booting
L (Default)	L (Default)	Only UFS boot
L (by external PU)	H (by external PU)	Only eMMC boot
H (by external PU)	L	Reserved
H (by external PU)	H (by external PU)	Reserved

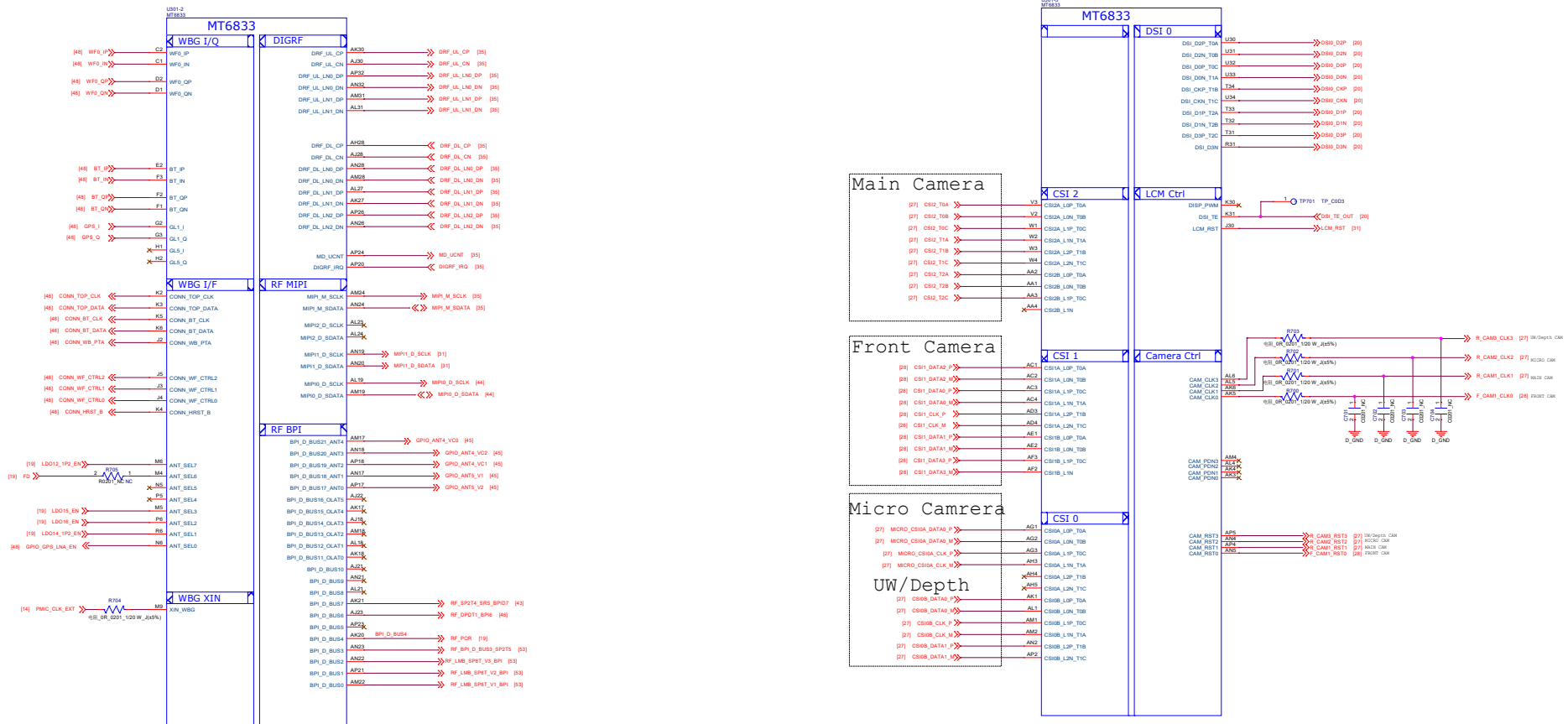
Note 13-5: "AUD_DAT_MOSI1" and "AUD_DAT_MOSI2" pin features in trapping pin to enable DDR.

AUD_DAT_MOSI1	AUD_DAT_MOSI2	SPMI_P_SCL	DDR
L (Default)	L (Default)	L (Default)	LP4x-MCP-4266Mbps
L (by external PU)	H (by external PU)	L (Default)	LP4x-Discrete-4266Mbps
H (by external PU)	L	L (Default)	Reserved
H (by external PU)	H (by external PU)	L (Default)	Reserved

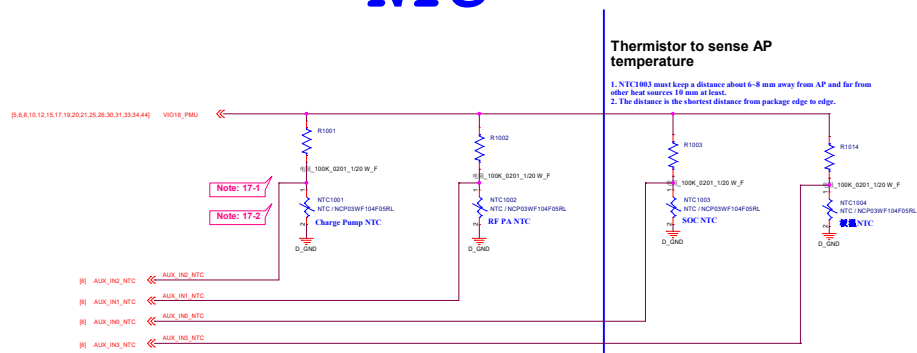
Note: SPMI_P_SCL for DDR 4266Mbps/3733Mbps Co-load, 'L' = 4266Mbps, 'H' (by external PU) = 3733Mbps

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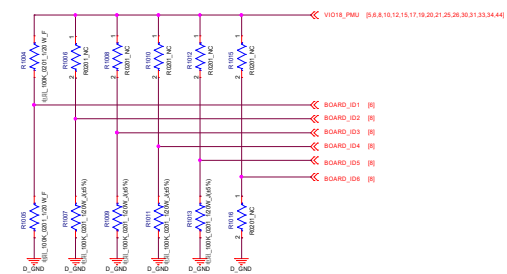
Title			06.BB_1
Size	Project	N17	
D		Rev	
		v1	
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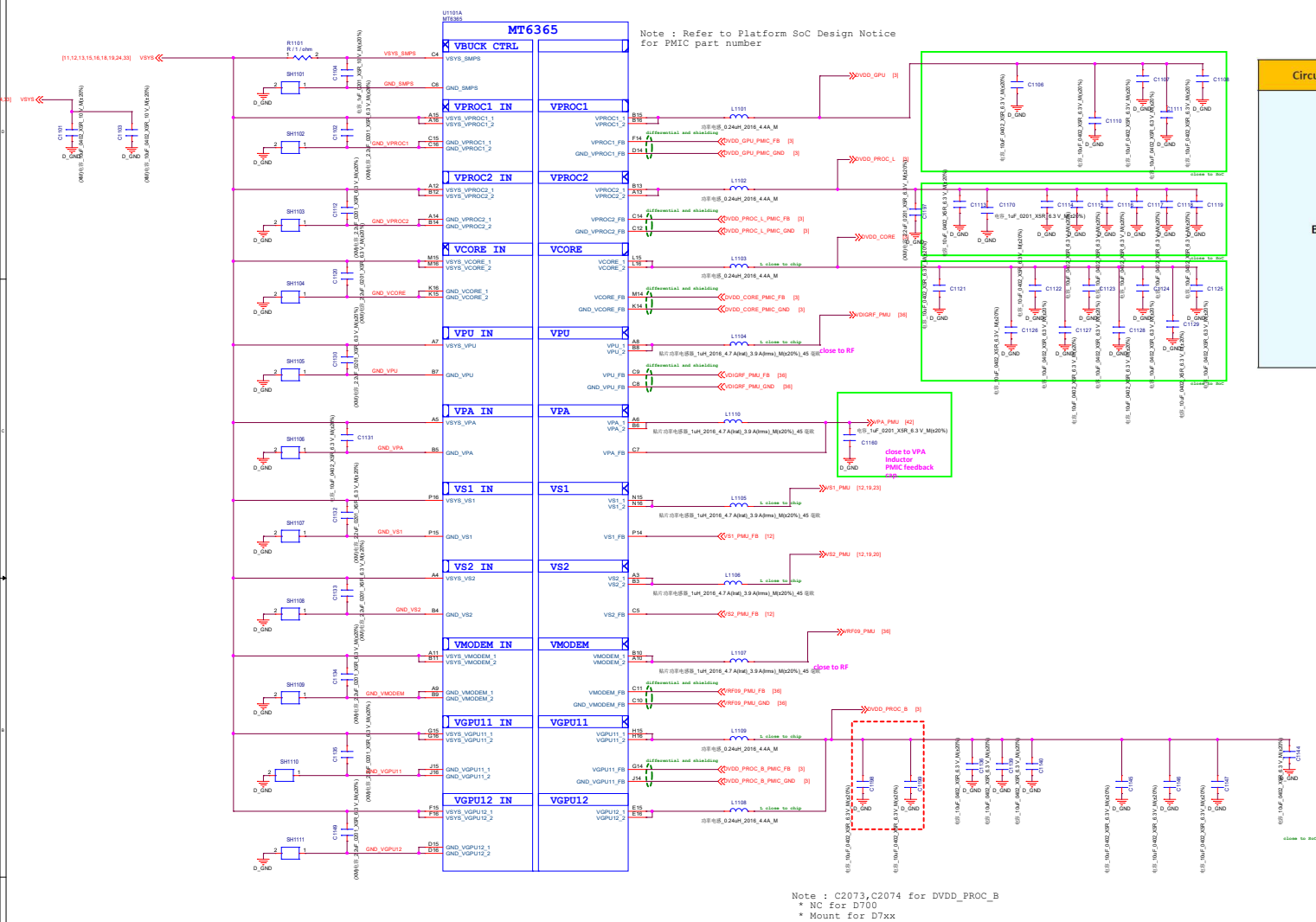
NTC



Board ID



华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title 10.BB_AUXADC_Thermal/BOARD_ID			
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Circuit Type	Buck Name (Application name)	Output Voltage Range (V)	Boot Default (V)	I _{OUT-MAX} (mA)
Buck	VPROC1 (DVDD_GPU)	0.40 ~ 1.19 (6.25mV/step)	ON (0.75)	4800
	VPROC2 (DVDD_PROC_L)	0.40 ~ 1.19 (6.25mV/step)	ON (0.75)	4800
	VGPU11+12 (DVDD_PROC_B)	0.40 ~ 1.19 (6.25mV/step)	ON (0.75)	4800 *2
	VCORE (DVDD_CORE)	0.40 ~ 1.3 (6.25mV/step)	ON (0.75)	4800
	VMODEM (VRF09_PMU)	0.50 ~ 1.10 (6.25mV/step)	OFF	4800
	VPU (VDIGRF_PMU)	0.40 ~ 1.19 (12.5mV/step)	ON (0.7)	2400
	VS1 (VS1_PMU)	1.86 ~ 2.20 (12.5mV/step)	ON (2.0)	2200
	VS2 (VS2_PMU)	1.20 ~ 1.50 (12.5mV/step)	ON (1.35)	2500
	VPA (VPA_PMU)	0.50 ~ 3.40 (50mV/step)	OFF	1000

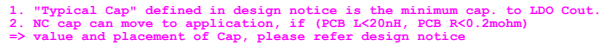
华勤通讯 Huaqin Telecom Technology Com.,Ltd

Title 11.POWER_MT6365_Buck

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MT6365



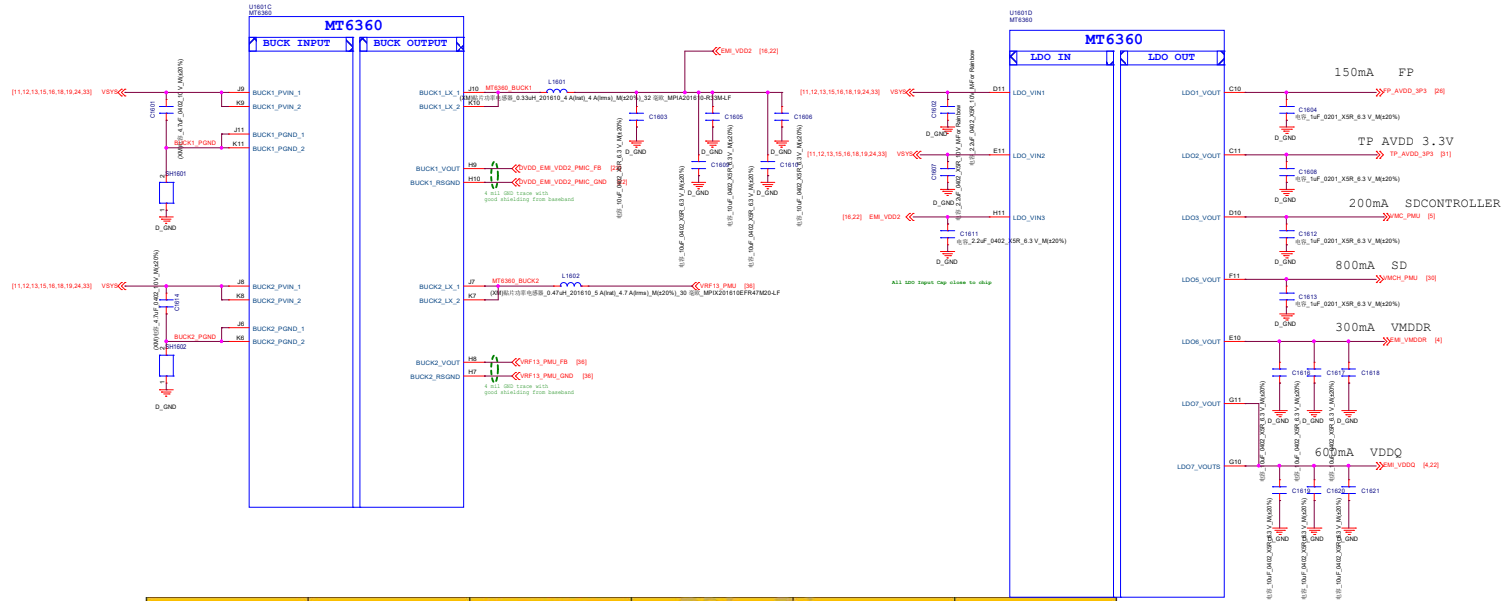
Circuit Type	LDOName	Output Voltage (V)	Boot Default (V)	Iboot (mA)	Expected use
DIDO	VSIM1	1.71/8.1/86/ 2.76/3.0/3.1	OFF (1.86)	200	SIM
	VSIM2	1.71/8.1/86/ 2.76/3.0/3.1	OFF (1.86)	200	SIM
	VCN33_1	3.3/3.4/3.5/3.6	OFF (3.3)	800	Connectivity
	VCN33_2	3.3/3.4/3.5/3.6	OFF (3.3)	800	Connectivity
	VMEM	2.55/2.9/3.0/3.3	ON (3.0)	Iboot	eMMC / UFS
	VI028	2.8/2.9/3.0/ 3.1/3.2/3.3	OFF (2.8)	200	IO & Sensor
	VIBR	2.7/2.8/3.3/3	OFF (2.8)	200	Vibrator

Circuit Type	LDO Name	Output Voltage (V)	Reset Default (V)	I _{OUT-MAX} (mA)	Expected use
SLD02	VA09	0.85	ON (0.85)	300	DIGIP SRAM (VRFOPS_MEM_PPMU)
	VA12	1.2	ON (1.2)	300	AP Analog Module
	VCN13	1.3	OFF (1.3)	350	Connectivity
	VSRAM_PROCI	0.60 ~ 1.2 (6.25mV/step)	ON (0.85)	600	CPUB SRAM
	VSRAM_PROCI2	0.60 ~ 1.2 (6.25mV/step)	ON (0.85)	600	CPUL SRAM
	VSRAM_OTHERS	0.60 ~ 1.2 (6.25mV/step)	ON (0.75)	600	OTHERS SRAM
	VSRAM_MD	0.60 ~ 1.2 (6.25mV/step)	ON (0.85)	600	GPU SRAM
	VRF12	1.2	ON (1.2)	800	AP Analog Module

Title	12.POWER_MT6365_LDO
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Note : Refer to Platform SoC Design Notice for PMIC part number



LDO1 VFP
LDO2 VTP
LDO3 VMC
LDO5 VMCH
LDO6 EMI_VMDDR
LDO7 EMI_VDDQ

Circuit Type	Name	Output Voltage Range (V)	Boot Default (V)	I _{OUT-MAX} (mA)	Expected use
Buck	BUCK1	0.3 ~ 1.3 (5mV/step)	ON (1.125)	3000	EMI_VDD2
	BUCK2	0.3 ~ 1.3 (5mV/step)	OFF (1.3)	3000	VRF13_PMU

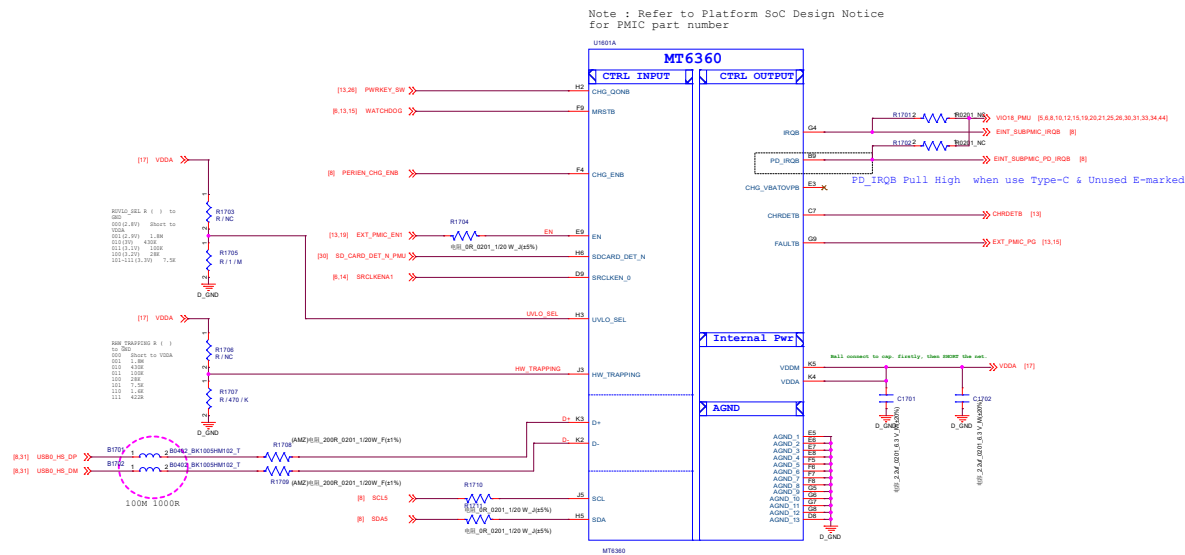
Circuit Type	Name	Output Voltage (V)	Boot Default (V)	I _{OUT-MAX} (mA)	Expected use
LDO	LDO1	1.8/2.0/2.1/2.5/2.7/ 2.8/2.9/3.0/3.1/3.3	OFF (1.8)	150	Fingerprint
	LDO2	1.8/2.0/2.1/2.5/2.7/ 2.8/2.9/3.0/3.1/3.3	OFF (1.8)	200	Touch Panel
	LDO3	1.8/2.9/2.8/3.0/3.3	OFF (3.0)	200	SD Card
	LDO5	2.9/3.0/3.3	OFF (3.0)	800	SD Card
	LDO6	0.75	ON (0.75)	300	EMI_VMDDR
	LDO7	0.6	ON (0.6)	600	EMI_VDDQ

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Title 16.POWER_MT6360_BUCK_LDO*

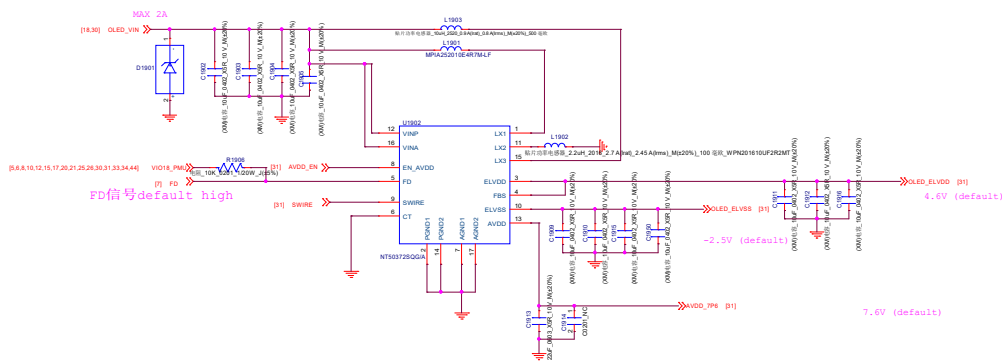
Size D	Project N17	Rev v1
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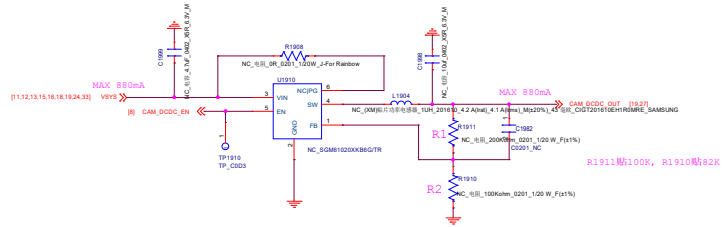


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Title 17.POWER_MT6360_General		
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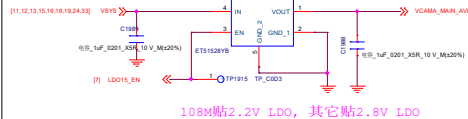
OLED Driver



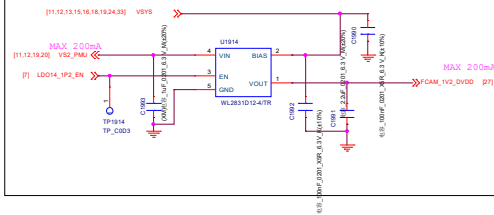
DCDC (108M CAM Only)



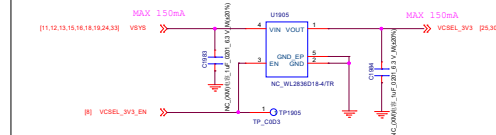
主摄AVDD 2.8V/2.2V LDO 预留NC



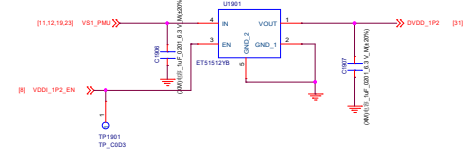
50M DVDD 1.2V LDO 预留NC



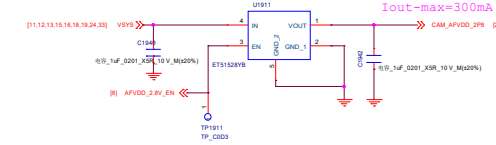
VCSEL 3.3V LDO 预留NC



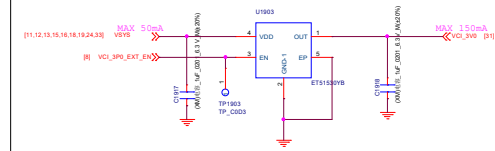
OLED VDDI 1.2V LDO



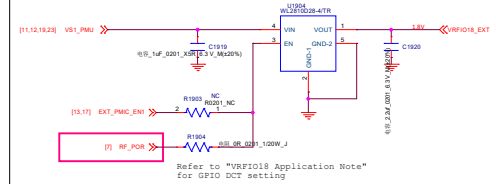
CAM AFVDD 2.8V LDO



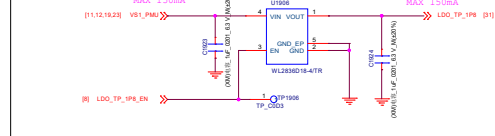
OLED VCI 3V LDO



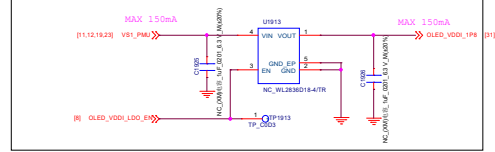
RF POWER 1.8V LDO



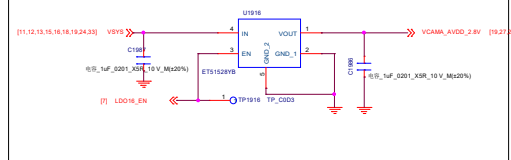
TP 1.8V LDO



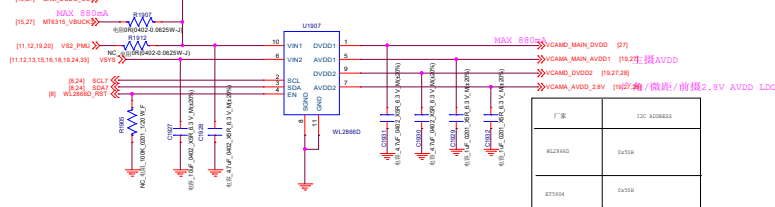
OLED VDDI 1.8V LDO 预留NC



广角/微距/前摄2.8V AVDD LDO 预留NC

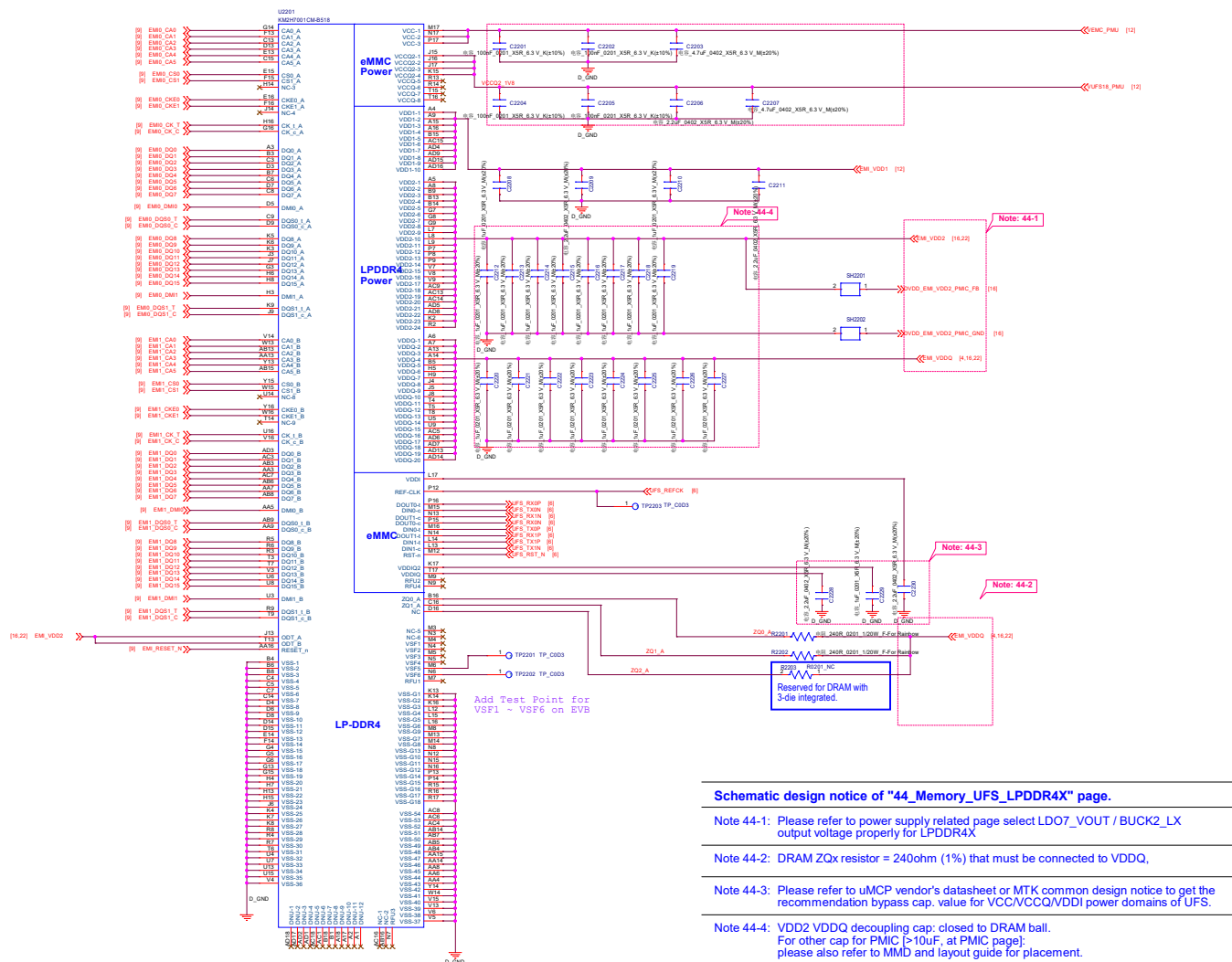


4 Channel CAM LDO

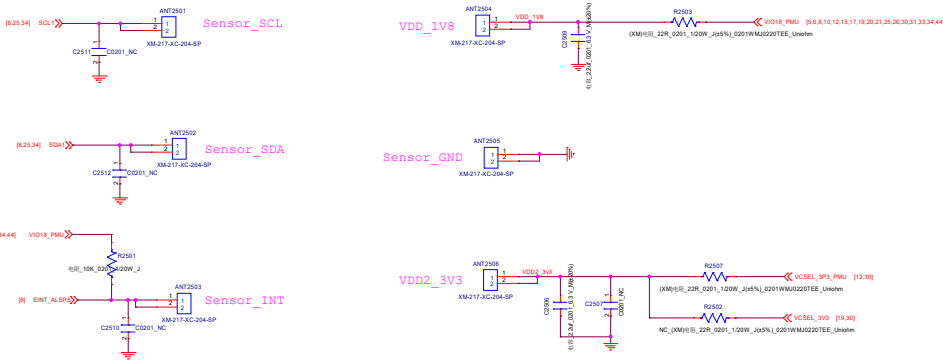
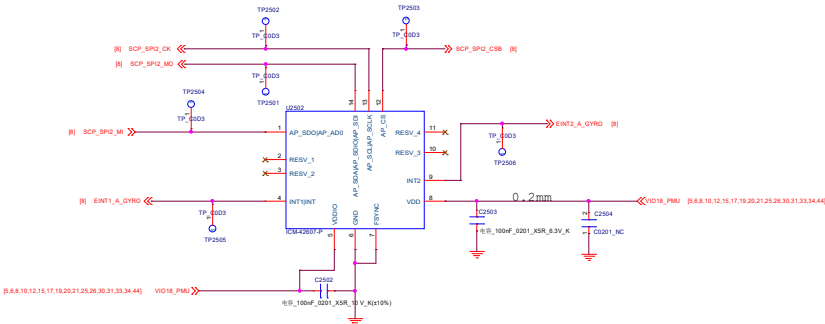
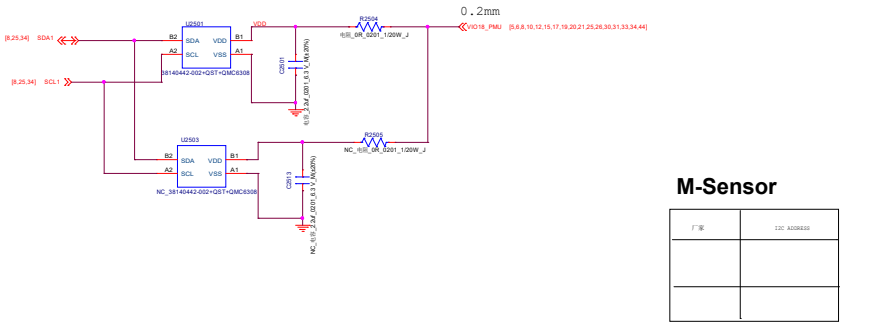


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Title 22.Memory_UFS_LPDDR4X		
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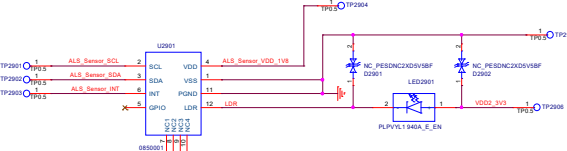


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Title25.SENSOR		
SizeD	ProjectN17	RevV1
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[illegible]

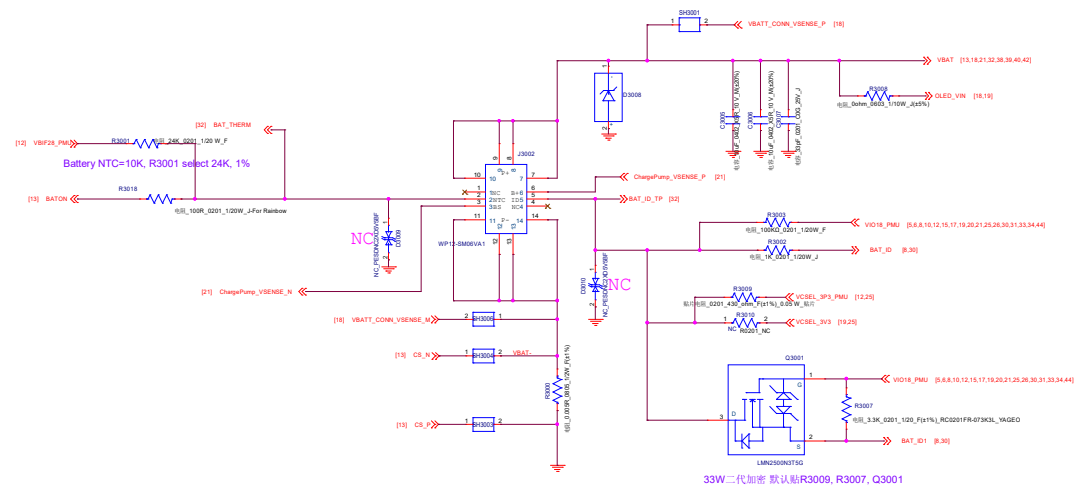
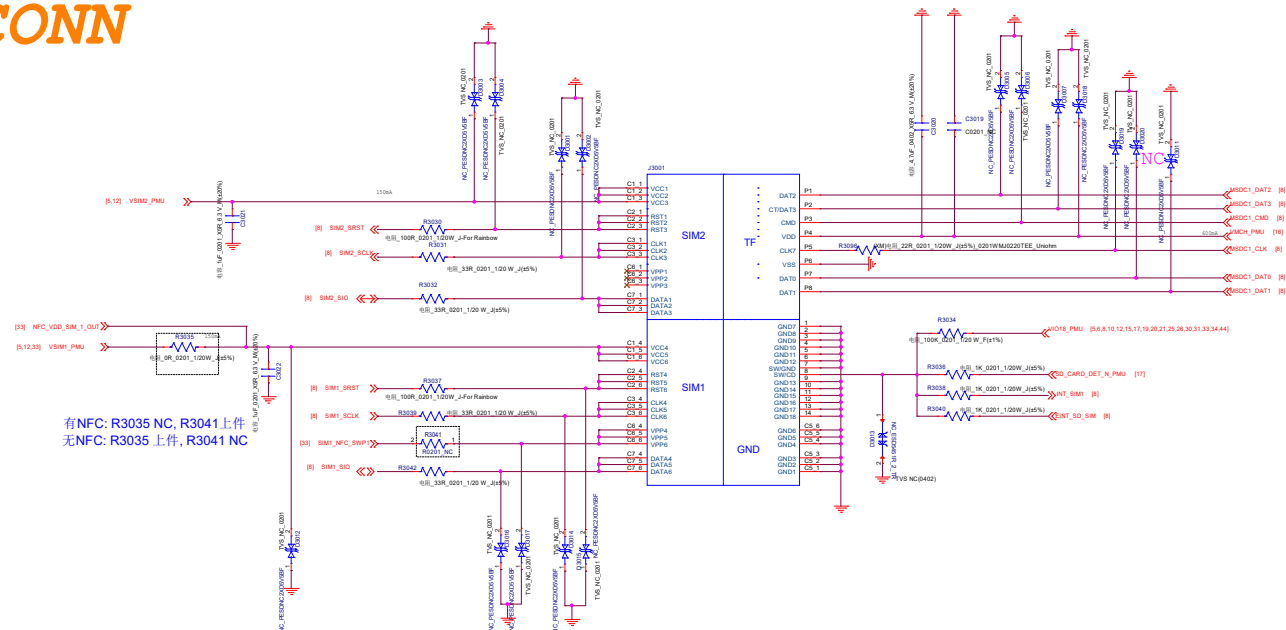
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Title 28.FRONT_CAM			
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ALS Sensor & VCSEL



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Title SENSOR_2		
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BATT CONN



华勤通讯 Huaqin Telecom Technology Com.,Ltd

Title	
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30.SIM/BATT

Size	D
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	Project
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[illegible]

Volume Down TP3204 1 TP_SMT_C10D_5 VOLUME_VOL_N (2H)

Volume Up TP3205 1 TP_SMT_C10D_5 VOL_P (2H)

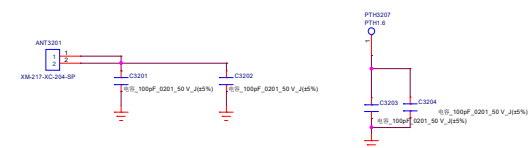
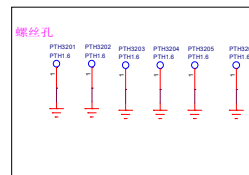
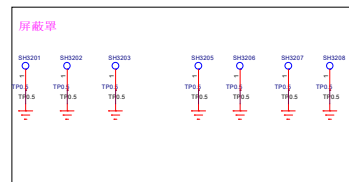
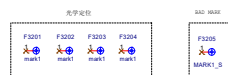
Power On TP3202 1 TP_FATP_C10D_5 PHONE_ON (2H)

GND TP3214 1 TP_FATP_C10D_5

Figure 1 shows four TP constructs: TP207 (USB_TP_CC), TP208 (USB_VBUS_CONN), TP209 (USB0_TP_DM), and TP2011 (USB0_TP_DP). Each construct is represented by a horizontal line with a red arrow pointing right, a blue circle with a white '1' inside, and a red arrow pointing left. The constructs are labeled on the left and right sides.

RX TP_C0D8 TP3210 1 中国 1K 0201 1/20W J R3201 URXD0 [8]

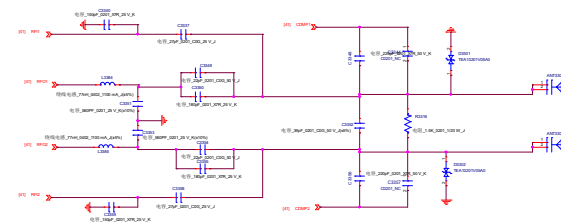
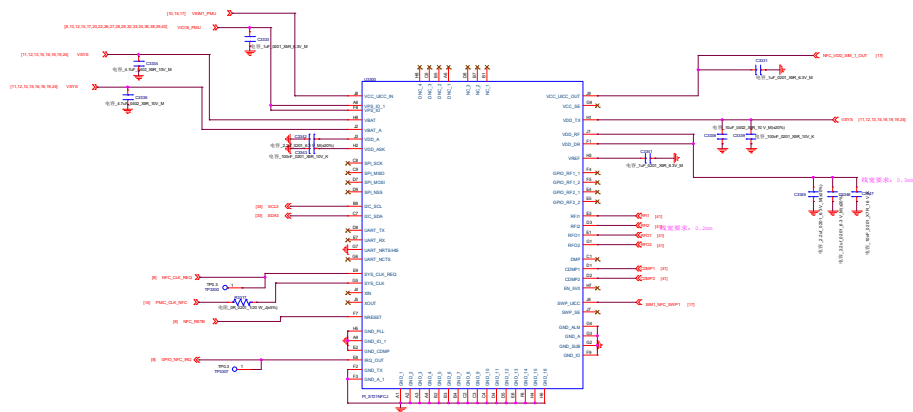
TX TP_C0D8 TP3212 1 中国 1K 0201 1/20W J R3202 UTXD0 [8]



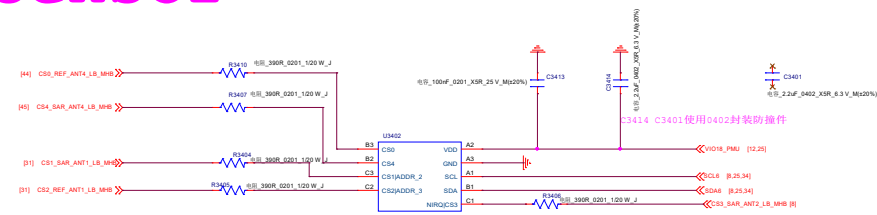
华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title 32.TEST_POINT/SHIELD			
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国内不做NFC

RF_NFC_ST21NFCJ

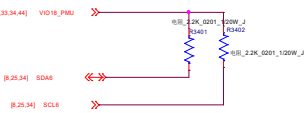


SAR_sensor



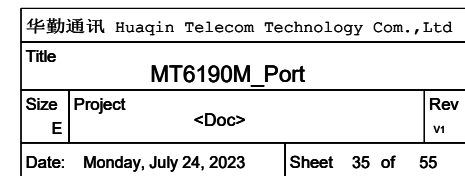
ANT2没有参考通道，SAR_sensor摆放时请尽量靠近ANT2的馈点，感应通道走线越短越好。

Design notice:
1) CS0_1 nets are differential pairs;
2) The width for each line, should be less than 0.1mm;
3) 0.2mm should be guaranteed between the ground and differential pairs;
4) And also, the distance between the differential line should be more than 0.2mm;
5) All the CSx nets should be protected by surrounding GND;

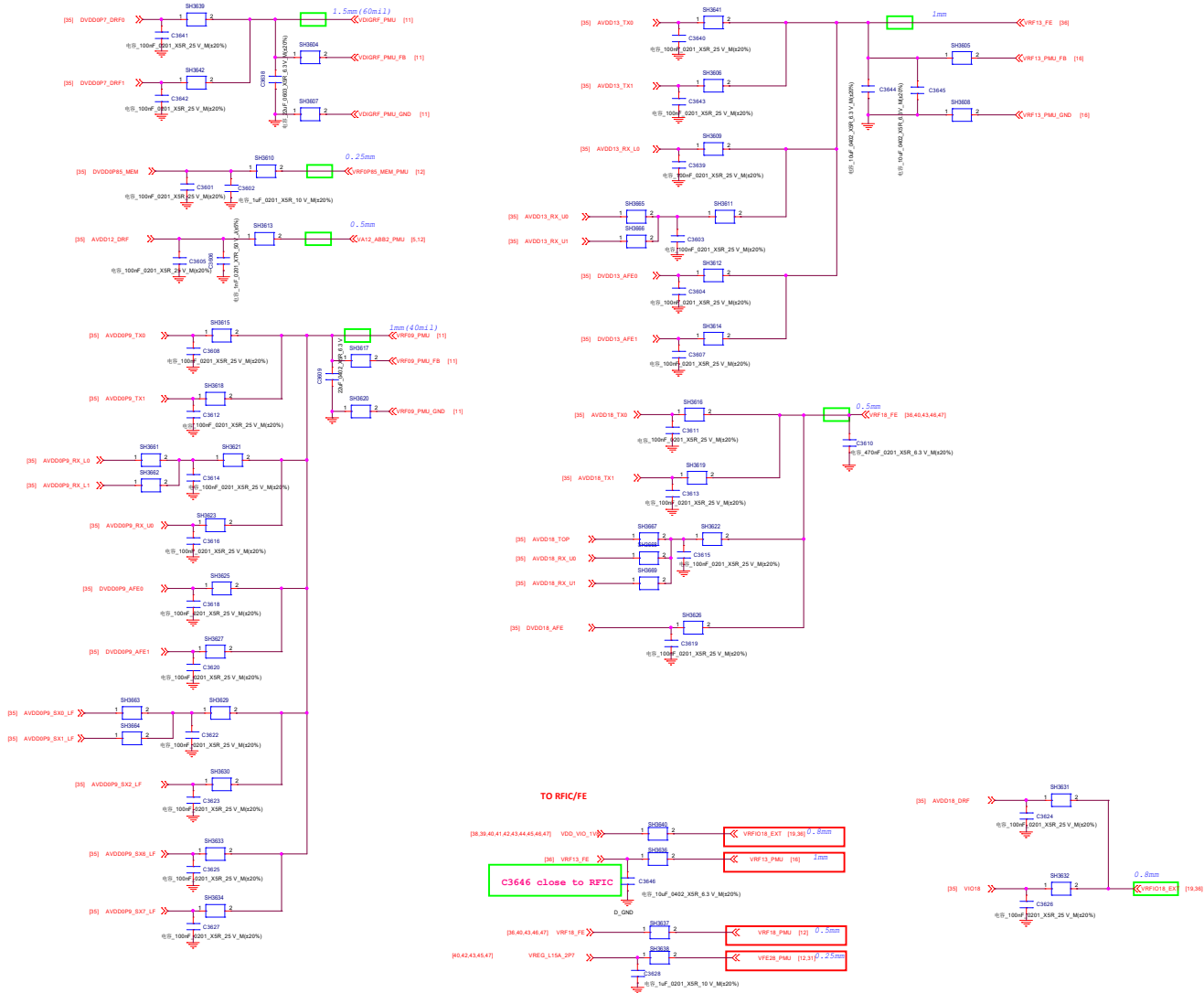


国内不做SAR_sensor，天线要求走线预留，BOM中NC

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Title SAR_sensor		
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MT6190M_PWR



华勤通讯 Huaqin Telecom Technology Com.,Ltd

Title

MT6190M_PWR

Size D	Project
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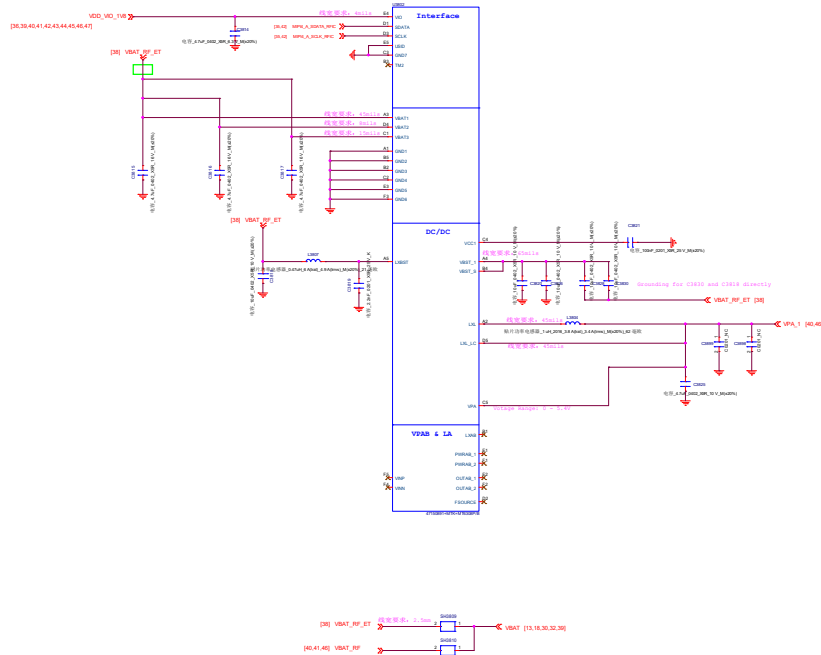
<Doc>

Rev
V1

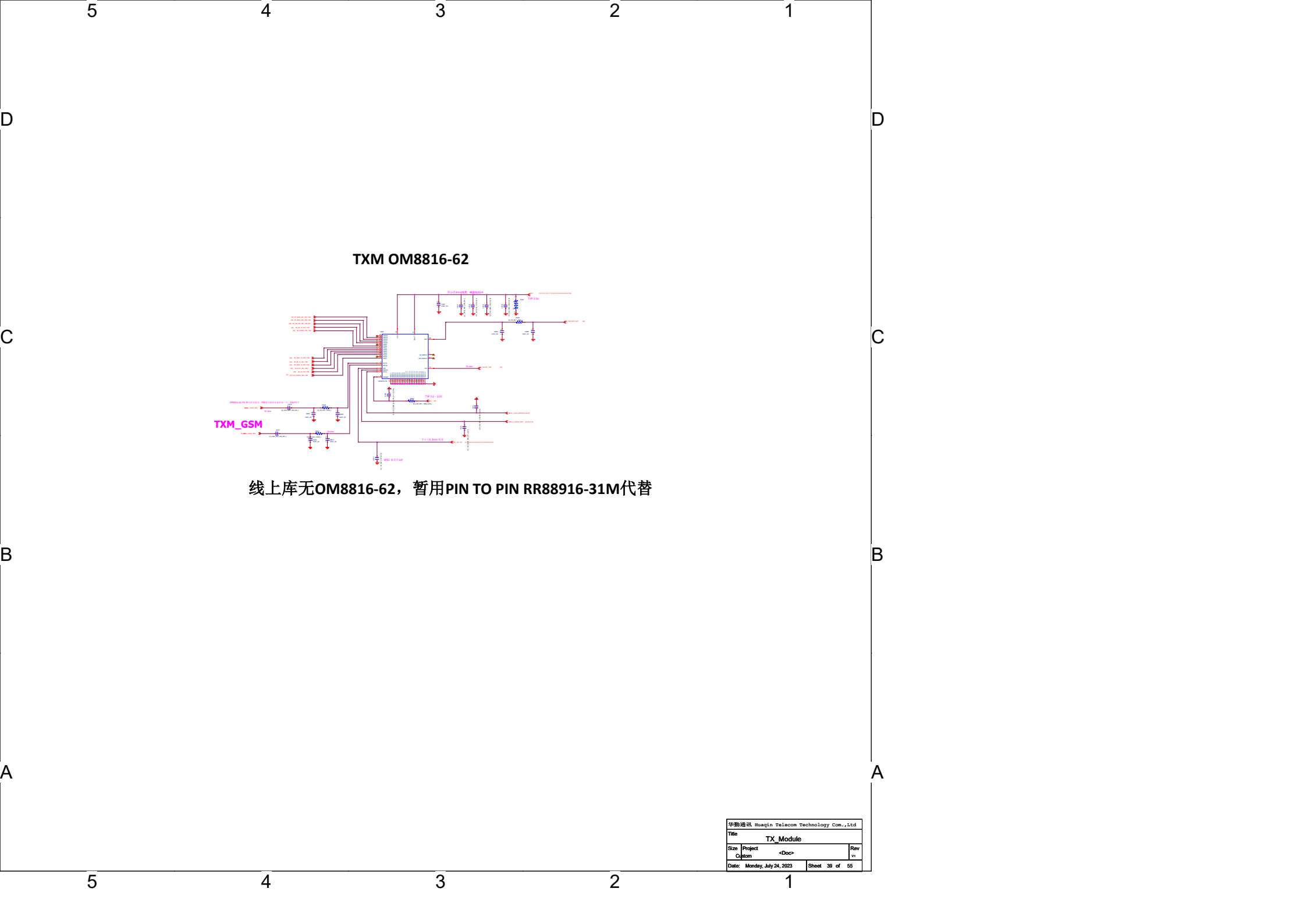
Date: Monday, July 24, 2023

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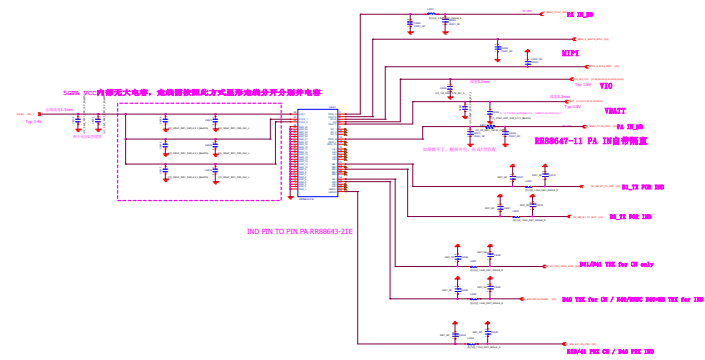
MT6308 FOR Phase 5N PA_N77_PA



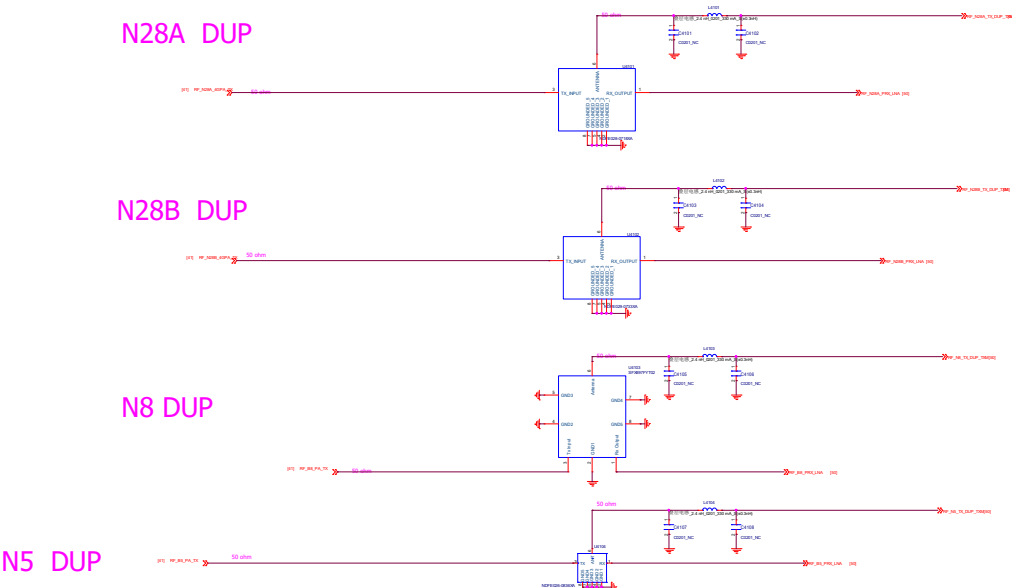
华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title			
RF_APT_MT6308M			
Size E	Project <Doc>		Rev v1
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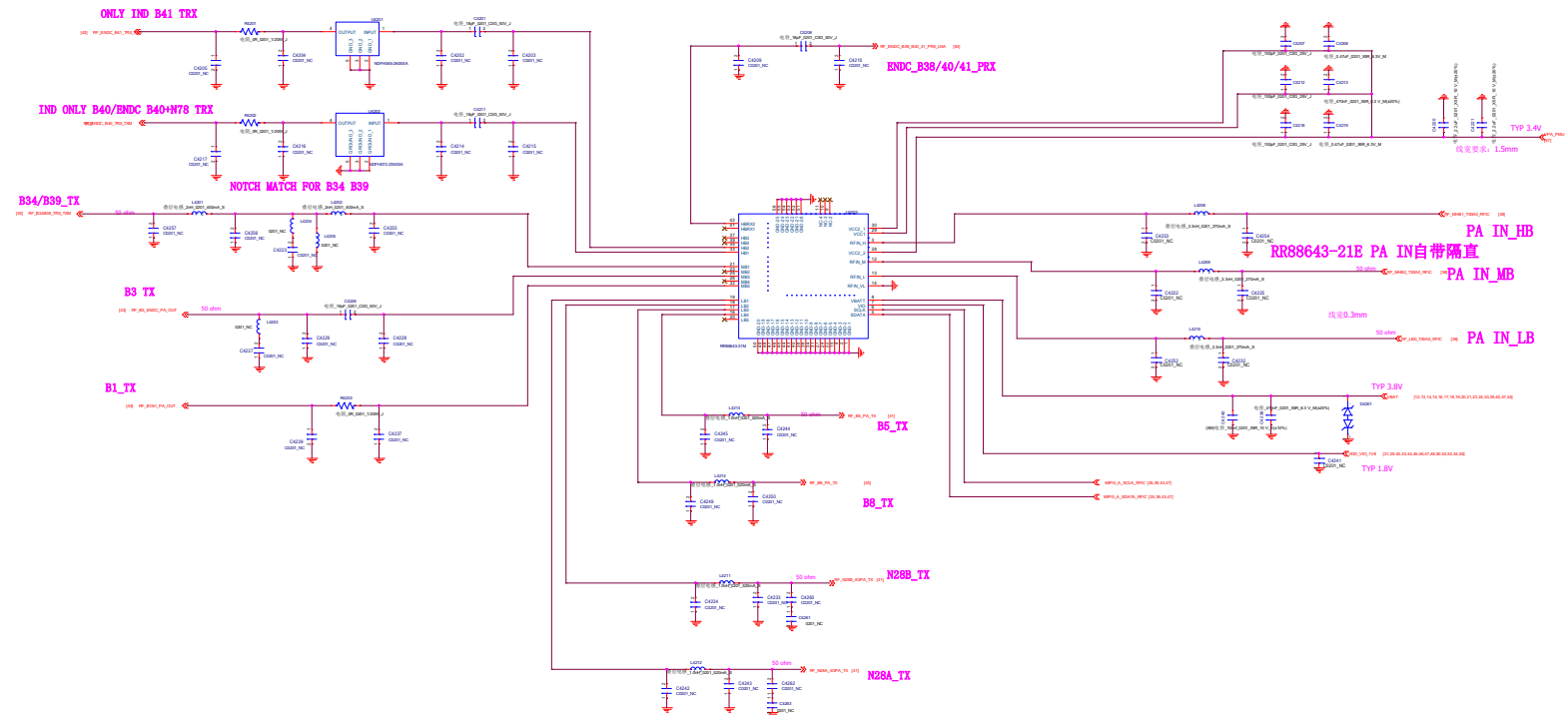
TDD PA RR88647-11



TRX_LB_DUP



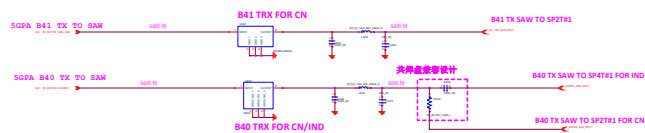
RR88643-21E_MMMB PA_LMHB



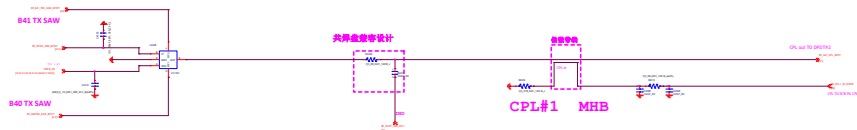
线上库无RR88643-21E，暂用PIN TO PINRR88643-31M代替

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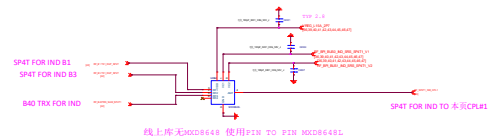
TDD PA TDD TRX SAW



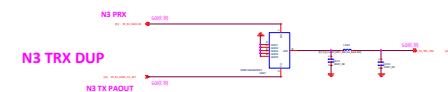
SP2T#1 ONLY CN B40/41 TRX



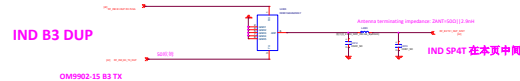
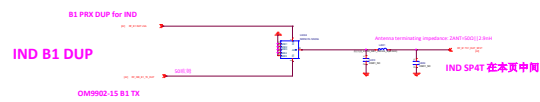
SP4T#1 FOR IND B1/B3/B40 TRX



N1/N3 TRX DUP

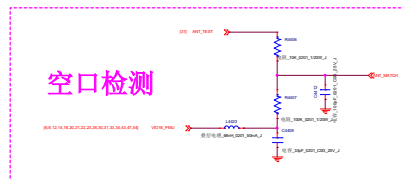
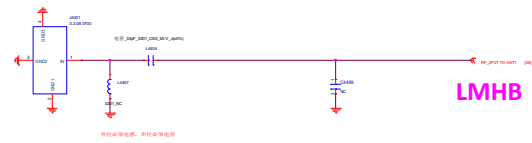


B1/B3 DUP for IND



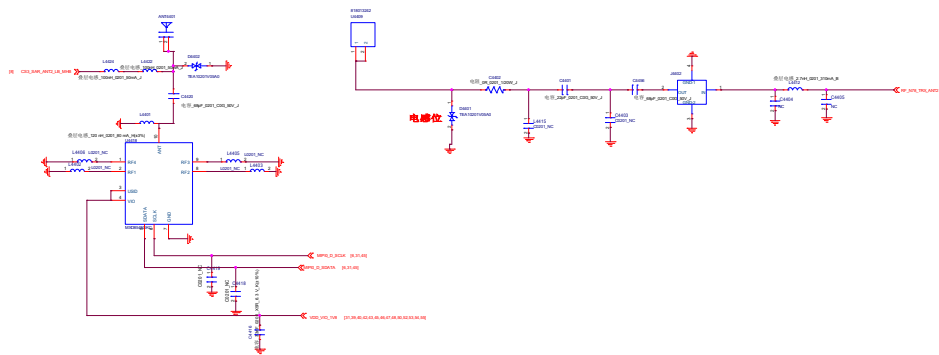
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ANT1 (LB_MHB_P) to 小板

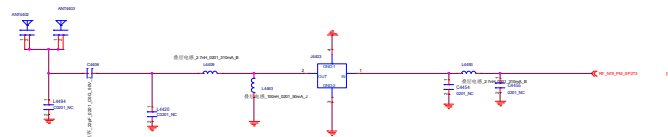


所有测试座到天线的电容 都需50v耐压

ANT2 (N78_PRX)

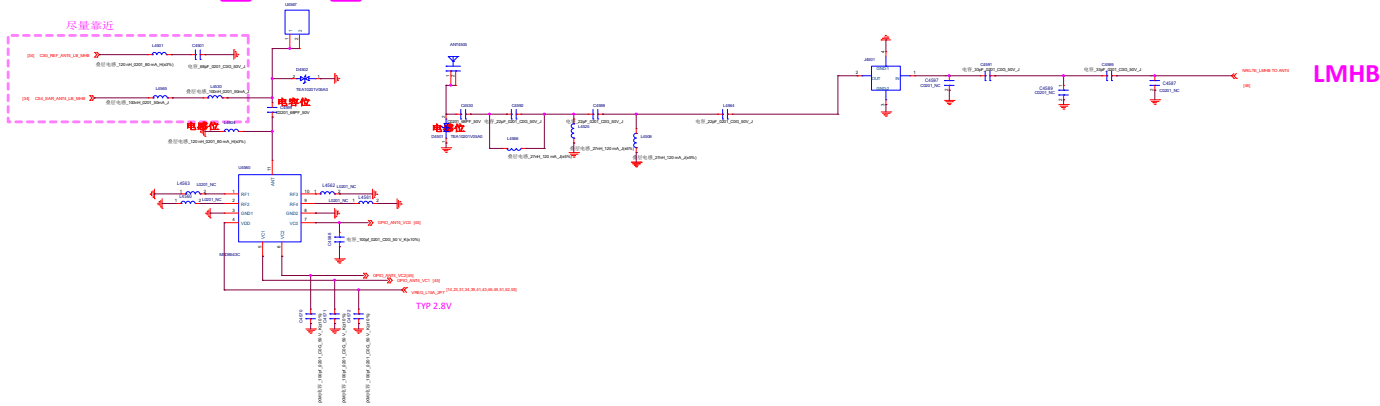


ANT3 (N78_M)

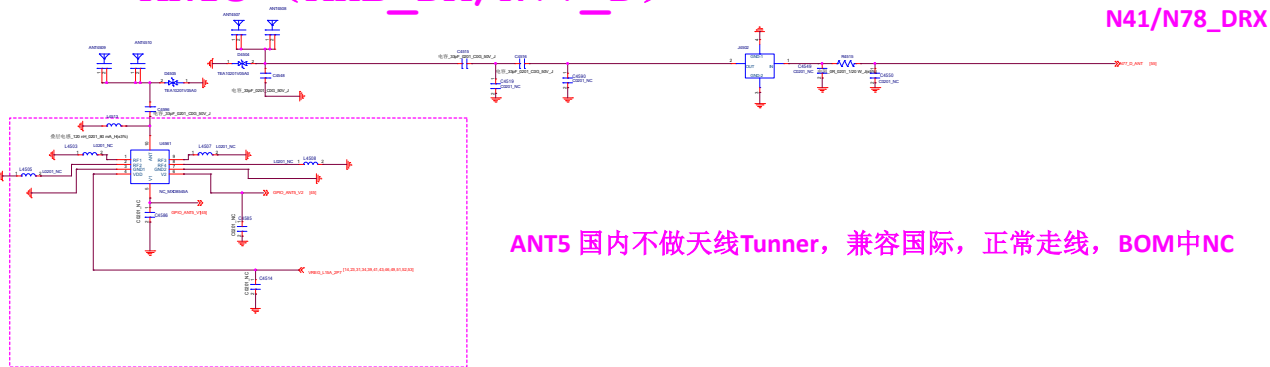


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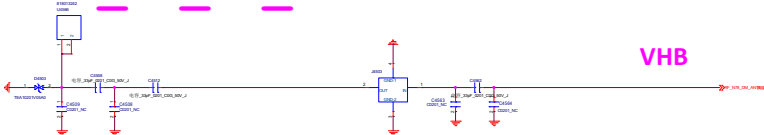
ANT4 (LB_MHB_P)



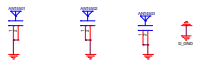
ANT5 (MHB_DM/N77_D)



ANT6 (N77_DM_HB_DRX)

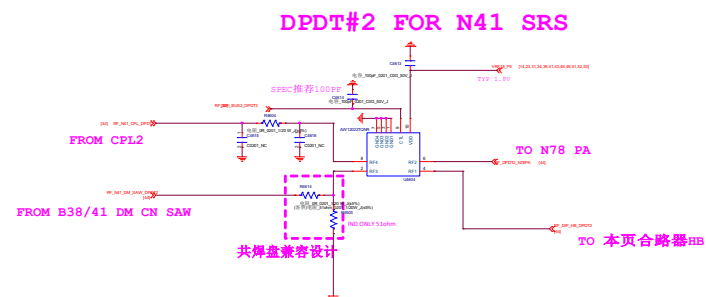
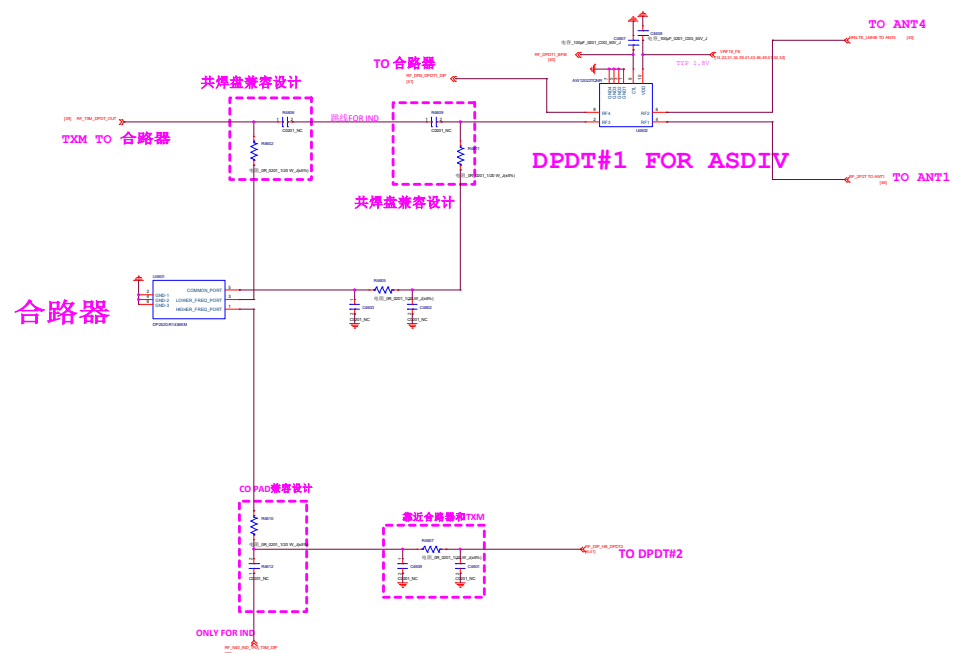


接地弹片



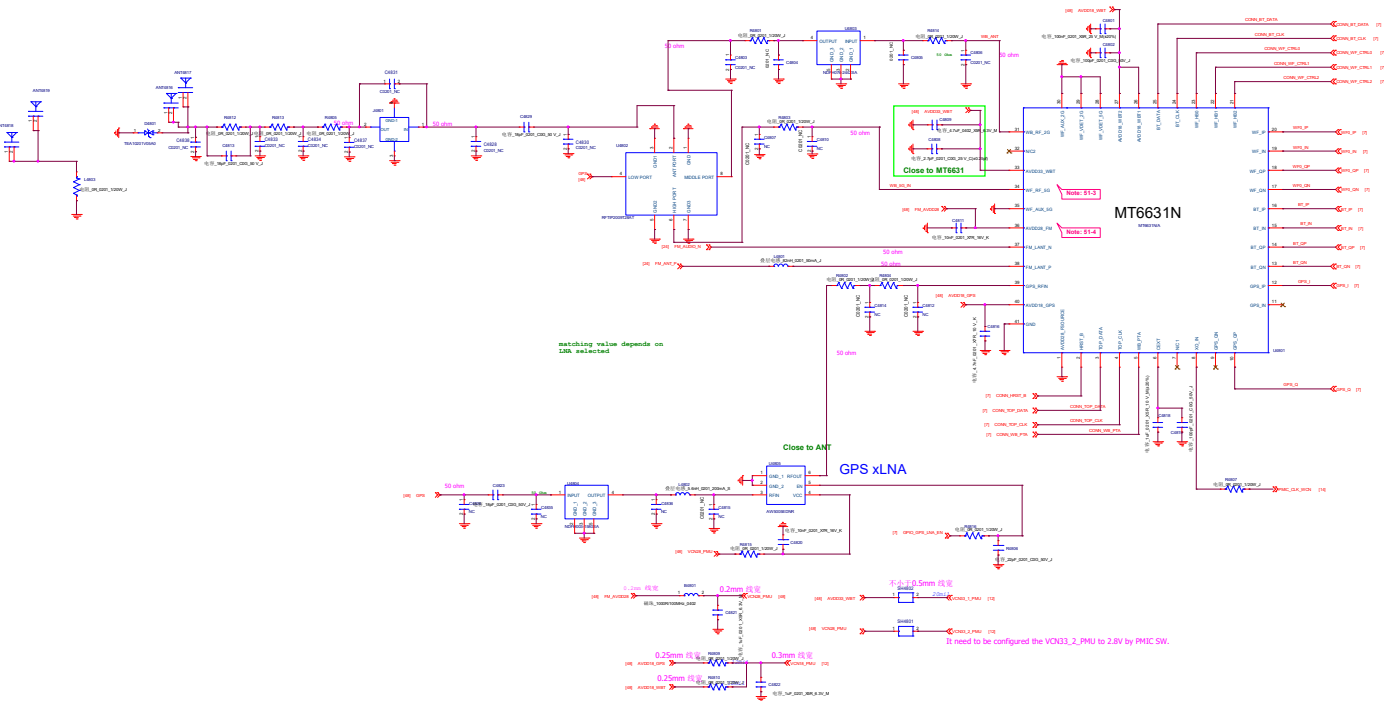
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2 DPDT



ANT7
WiFi 2.4G & 5G_GPS

MT6631



Schematic design notice of "51_CONNECTIVITY_CONSYS_MT6631"

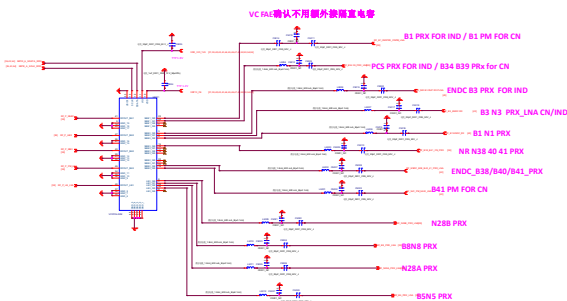
Note 51-1: For R6015 size, please select 0402 size or larger one

Note 51-2: Please refer to MT6765 Baseband design notice for VCN33_LDO selection guide

Note 51-3: If WiFi 5G not support, connect pin 34(WF_RF_SG) to GND

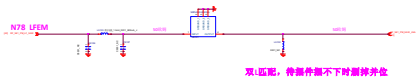
Note 51-4: Pin 36 (AVDD28_FM) must be connected to VCN28 even if FM not support

PRX_LNA BANK_VC3725-22M

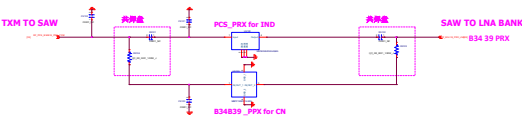


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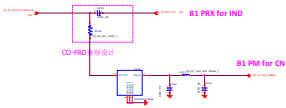
TDD B41 PM RX SAW FOR CN



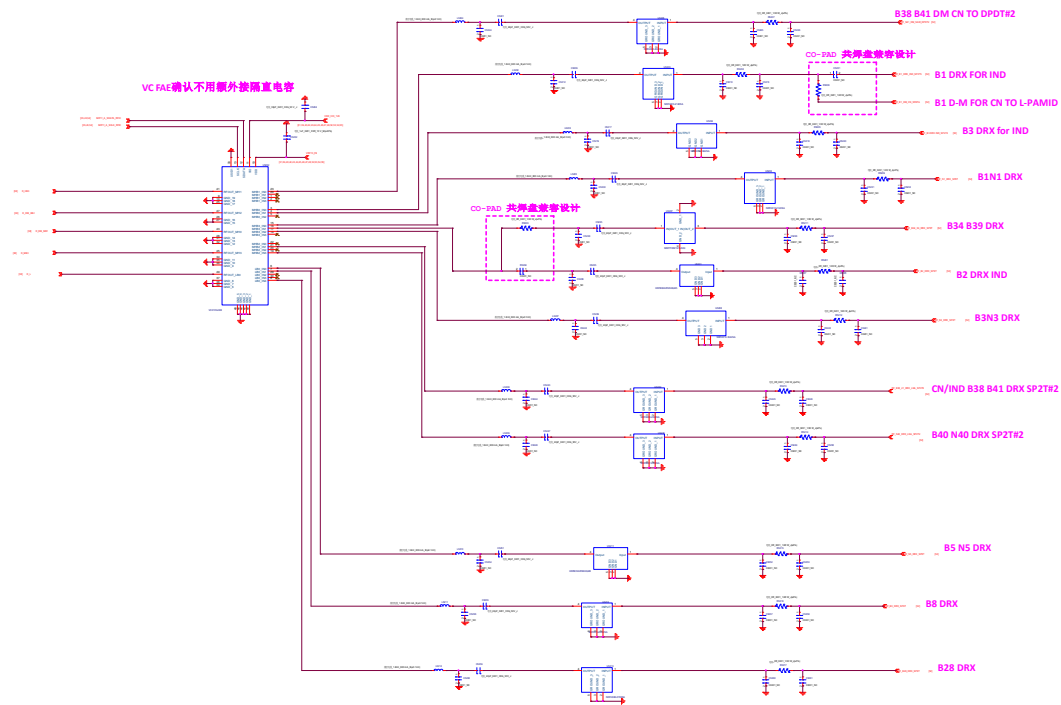
B34/39 PRX PCS PRX SAW



B1 PRX FOR IND / B1 PM FOR CN

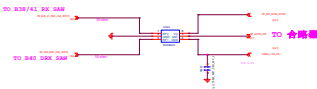
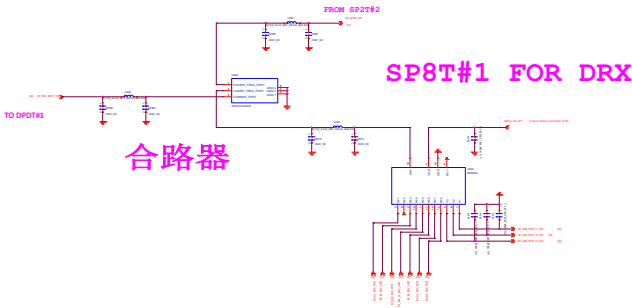
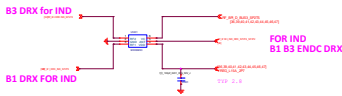


DRX_LNA BANK_VC3725-22M



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SP2T#5 FOR B1/B3 DRX IND ONLY

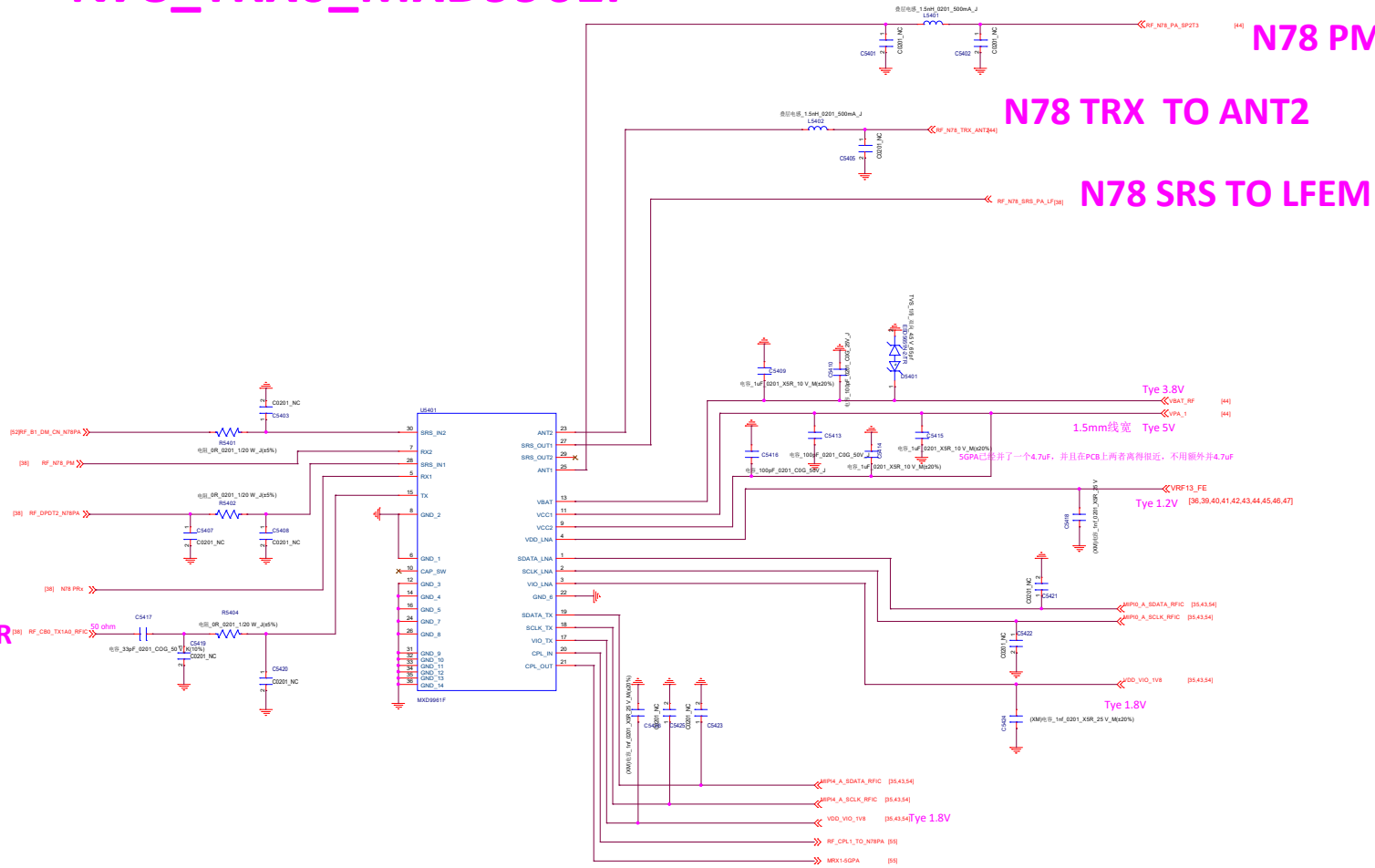


SP2T#2 FOR B38/41 DM & B41 DRX

N78_TRX0_MXD9961F

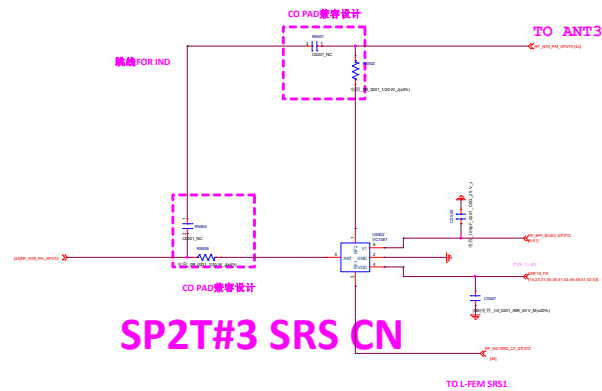
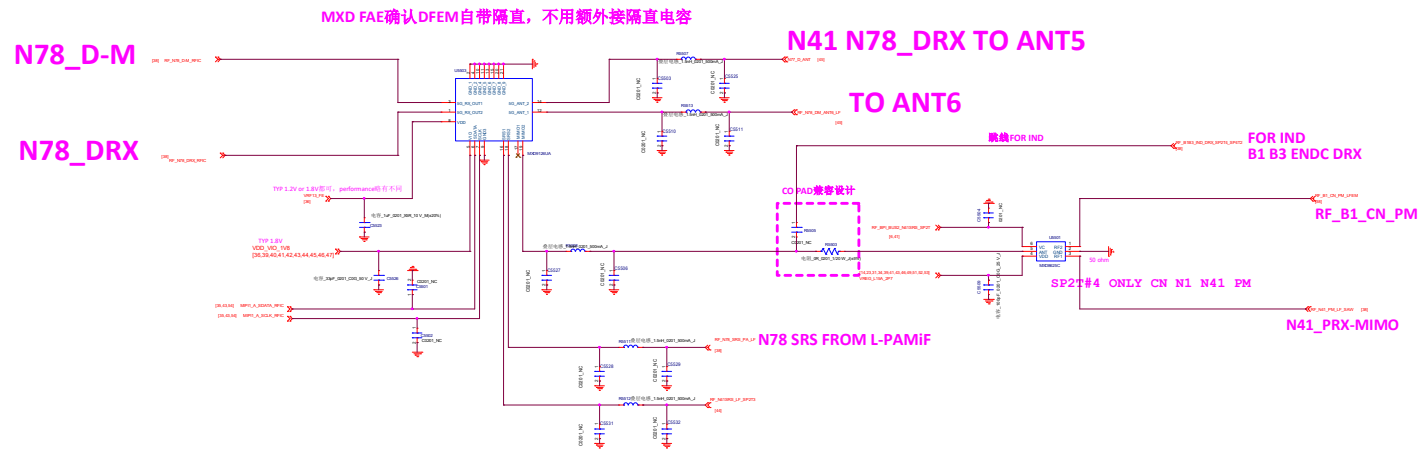
B1 D-M FOR CN
N78 PM TO WTR
FROM DPDT#2

N78 PRX TO WTR
N78 TX FROM WTR



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N78_LFEM_MXD9126UA



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